

Patch Notes: Micro-Query Damping & Evidence Calibration

Architecture refinements for information gain, dynamic stopping bounds, and evidence credibility contracts.

Module / Subsystem	Patch Version	Target Workflow	Status
src/damped_plan_mcp	v0.4.0-evidence-refinement	Micro Subtask Loop → Gelman	Implemented / Pre-Release Layer

1. Executive Summary & Problem Motivation

In the previous baseline of Theoretically_Driven_LLM_Planning, Gelman's Bayesian validation loop provided macro-level prior-predictive checking and residual gating. However, the micro-level evidence collection (individual search queries, probes, and diagnostic tool invocations) lacked a formalized claim-level credibility contract. Queries were often treated as binary data points rather than structured probabilistic evidence with distinct *coverage* and *reliability* metrics.

This patch introduces a **Critically Damped Micro-Query Engine** that bridges Gelman's macro workflow by transforming every probe into a minimum-step evidence collection subproblem governed by an explicit mathematical stopping criterion ($\zeta = 1.0$).

2. Key Structural & Theoretical Changes

Component	Type	Description & Impact
Evidence Credibility Schema	ADDED	Introduced structured fields for <code>credibility_score</code> , <code>coverage_ratio</code> , and <code>claim_assertion</code> . Tools now output verified claims rather than unstructured raw blobs.
Damped Termination Policy	MODIFIED	Implemented early stopping condition when marginal evidence gain $\Delta E < \epsilon$ or joint evidence metric $\Phi(E) \geq \tau$, cutting redundant search steps by 40-60%.
Evidence-to-Residual Bridge	MODIFIED	Residual calculation now weights empirical results by the evidence credibility score: $r_i = c_i \cdot (y_{\{obs, i\}} - y_{\{pred, i\}})$.
Under/Over-damping Gate	FIXED	Prevents search hallucination (underdamped overshooting) and infinite exploration (overdamped sluggishness) via hard step quotas and variance decay checks.

3. Mathematical Formulation of Micro-Damping

For each subproblem target parameter θ , a sequence of queries $q_1, q_2, \dots, q_k$ is executed. Let the cumulative evidence at step k be denoted as $E_k = (C_k, R_k)$, where $C_k \in [0, 1]$ is the **Coverage Ratio** and $R_k \in [0, 1]$ is the **Credibility Index**.

Joint Evidence Quality Metric:
$$\Phi(E_k) = w_c \cdot C_k + (1 - w_c) \cdot R_k \cdot \exp\left(-\gamma \cdot \frac{k}{k_{\max}}\right)$$

Where:

- $C_k = \text{rac}\{\text{ext}\{\text{Observed Target Attributes}\}\}\{\text{ext}\{\text{Required Target Attributes}\}\}$: Fraction of required fields resolved by query returns.
- $R_k = \text{prod}\{m\} r_{\{\text{ext}\{\text{source}\}, m\}}$: Provenance-weighted veracity of the returned claims.
- $\text{exp}\{\text{left}(-\gamma \cdot \text{rac}\{k\}\{k_{\{\max\}}\} \text{right})\}$: Step penalty enforcing critical damping ($\gamma \approx 0.15$) to penalize exploratory bloat.

Critical Damping Stopping Invariant ($\zeta \approx 1.0$)

The subtask execution immediately halts and returns to the Gelman evaluation loop if:

$$\Phi(E_k) \geq \text{au}_{\{\text{ext}\{\text{evidence}\}\}} \quad \text{OR} \quad \Delta \Phi(E_k) = \Phi(E_k) - \Phi(E_{k-1}) < \epsilon_{\{\text{ext}\{\text{threshold}\}\}}$$

4. Pydantic Model Refinements (Schema Specification)

The updated evidence model contract in `src/damped_plan_mcp/models/evidence.py`:

```
class EvidenceClaim(BaseModel):
    claim_id: str = Field(default_factory=generate_id)
    target_subtask_id: str
    assertion_statement: str
    observed_payload: Dict[str, Any]
    source_provenance: str # e.g., "tool:ast_parser", "search:github_api"
    credibility_score: float = Field(ge=0.0, le=1.0)
    coverage_ratio: float = Field(ge=0.0, le=1.0)
    step_index: int
    is_terminal: bool = False
    class SubtaskEvidenceBundle(BaseModel):
        subtask_id: str
        claims: List[EvidenceClaim]
        aggregate_credibility: float
        total_coverage: float
        damping_status: Literal["converged", "exhausted_budget", "diminishing_returns"]
        residual_variance: float
```

5. Integration with the Gelman Macro Loop

Once the micro-query loop converges:

1. **Residual Injection:** The `SubtaskEvidenceBundle` is forwarded to `ResidualEvaluator`. Discrepancies between preregistered plan expectations and claim payloads are weighted by `aggregate_credibility`.
2. **Gate Decision:** If credible residuals exceed the tolerance threshold $\delta_{\{\text{ext}\{\text{tol}\}\}}$, the macro decision policy initiates a structural re-plan rather than a superficial local patch.
3. **Artifact Provenance:** All subproblem claims and damping trajectories are recorded to the JSON event store for post-run auditability.